

Phase B5 Synthesis and Closure: How the Operational Test Architecture Completes the Series B Governance Specification

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

Abstract

This note is the fifteenth and final note of Phase B5, the fifth phase of the Series B derivation sequence derived from Paper 2 of the Coordination Knowledge Substrate (CKS) theory. It synthesizes Phase B5’s contribution to Series B, locates that contribution within the cumulative governance specification built across Phases B1 through B5, and previews Phase B6. Phase B5’s specific contribution is verifiability: where Phase B2 established how the twenty Paper 2 commitments operationally decompose, Phase B3 established what violations of those commitments look like, and Phase B4 established how commitments depend on each other, Phase B5 established how governance tests whether those commitments are correctly instantiated in a given deployment. The resulting test architecture — approximately forty-one specific test questions organized both by commitment and by governance timing — makes the Paper 2 architecture verifiable, not merely specifiable. At the close of Phase B5, the Series B prior-art record comprises 205 notes; combined with the 197 Series A notes, the complete derivation-note series has produced 402 notes of public prior art. Phase B6 will formalize the edges of the architecture — deployment configurations where Paper 2 commitments approach their limits — as approximately fifteen boundary case notes.

1. The question Phase B5 was built to answer

Each phase of Series B addresses a different governance question about Paper 2’s architecture. Phase B1 answered: what are the foundational commitments? Its twenty notes established that Paper 2 commits to the instinct/reasoning separation as independently-evolving layers, to three

architectural levels (cell, aspect, Self), to a DNA layer and action layer within every cell, to lifecycle governance primitives (birth, mating, death) at every level, to three evolution mechanisms in productive tension, to multi-shaped human governance across those mechanisms, and to the enterprise-brain Self as a coherent design pattern (§4–9 of Paper 2). Phase B2 answered: how do those commitments operationally decompose? Its 110 notes established the specific behavioral requirements entailed by each commitment — the moves governance must be capable of making if it instantiates the commitment at all. Phase B3 answered: what does violation look like? Its thirty notes established recognizable anti-pattern signatures for each commitment, giving governance an empirical vocabulary for detecting divergence. Phase B4 answered: how do commitments depend on each other? Its thirty notes established composition pairs — two-commitment interaction structures whose correct handling requires governance to hold both commitments simultaneously rather than each in isolation.

Phase B5 was built to answer the remaining question: how does governance *verify* that a Paper 2 deployment is correctly instantiated? Specification (B2), violation recognition (B3), and dependency mapping (B4) together tell governance what the architecture should look like, what failure looks like, and what breaks if a commitment is dropped. They do not, by themselves, provide a test protocol — a set of specific, answerable questions that a governance review can pose to a live deployment and obtain evidence about whether the architecture is present. Phase B5 supplies that protocol.

2. The test library Phase B5 produced

Phase B5 produced its test library across fourteen preparatory notes (B5.01–B5.14) before this synthesis note. The library has two organizational dimensions: by commitment and by governance timing.

Organization by commitment. Note B5.01 established the integrating frame: a Phase B5 test is a specific question addressable to a deployment whose answer provides evidence about whether a named Paper 2 commitment is correctly instantiated. The test is not a checklist of features; it is an evidential question whose answer either supports or undermines a governance claim. Notes B5.02 through B5.07 developed commitment-specific test batteries for six of the most architecturally central commitments: the instinct/reasoning separation itself, the three-level architecture (cell, aspect, Self), the DNA/action layer distinction within the cell, the expression mechanism as governed selection over DNA-layer activation, birth as governed origination, and mating as a governable primitive over content combination. Each of these six notes produced between five and eight specific test questions tied to the commitment's operational requirements as established in Phase B2.

Note B5.08 developed composition pair integrity tests: questions directed not at individual commitments but at the interaction points between two commitments that Phase B4 had established as architecturally coupled. A composition pair integrity test asks whether the constraints that each commitment places on the other are jointly satisfied — whether the

deployment's handling of the interaction is coherent from both commitments' perspectives simultaneously. Note B5.09 developed anti-pattern detection tests: questions that ask whether specific B3 anti-pattern signatures are present, using the violation vocabulary that Phase B3 established. Where B5.02–B5.07 ask whether a commitment is satisfied, B5.09 asks whether recognizable failure signatures are absent — a complementary evidential approach.

Across B5.02–B5.09, Phase B5 assembled approximately forty-one specific test questions organized by commitment and architectural dimension, covering the full set of Phase B1 foundational commitments with varying depth according to architectural centrality.

Organization by governance timing. Notes B5.10 through B5.13 reorganized the test library along a second axis: when in the deployment's lifecycle is governance in a position to apply each test? Note B5.10 assembled the initialization test suite — questions that can be answered (and must be answered affirmatively) at the moment a deployment is stood up, before any operational use. These questions establish baseline correctness: is the instinct/reasoning separation structurally present? Are the three levels defined with correct membership semantics? Are DNA and action layers distinct within every cell? Is expression implemented as governed selection? Note B5.11 assembled the ongoing governance test suite — questions addressable during regular operation that provide evidence of sustained instantiation rather than one-time setup. These questions test whether the commitments are behaviorally active, not merely architecturally defined. Note B5.12 assembled the evolution-event test suite — questions that governance should pose when a lifecycle event (birth, mating, death, instinct evolution, DNA evolution, action-feedback evolution, horizontal evolution, vertical evolution) occurs. Evolution events are the moments when the architecture is most likely to drift; B5.12 established the questions that confirm drift has not occurred. Note B5.13 assembled the compliance demonstration test suite — questions whose answers, taken together, constitute evidence suitable for a formal governance demonstration that the architecture is present, correctly instantiated, and operationally active.

Note B5.14 synthesized the full test battery: it assembled all four suites into a coverage map demonstrating that the forty-one test questions collectively cover every Phase B1 commitment across every governance timing, with no commitment left untested and no timing with an evidential gap.

3. Phase B5's verifiability contribution

The specific contribution Phase B5 makes to Series B is the addition of verifiability to what was already a specification. The distinction is important and worth naming precisely.

A governance specification is a set of commitments with operational content — a description of what an architecture that instantiates the commitments looks like, how it behaves, what it preserves, and what it refuses. Phase B2 produced the specification in full: its 110 notes establish the operational requirements of each commitment with enough precision that a designer can build a Paper 2-compliant architecture without ambiguity. A governance

specification is complete when every commitment is operationally decomposed. Phase B2 completed that work.

But a specification, however complete, does not by itself answer the governance question that arises after deployment: *is this deployment actually instantiating the commitments?* That question is not about design intent; it is about operational evidence. It requires a test — a question or procedure that produces observable evidence one way or the other. Phase B2's operational decompositions tell governance what to build; Phase B5's test library tells governance how to confirm what was built is what was intended.

Phase B3 and B4 made partial progress toward this confirmation function, but from different angles. Phase B3's anti-pattern library tells governance what to look for when something is wrong — the detectable signatures of commitment failure. Phase B4's composition pair library tells governance where the most architecturally consequential interdependencies are, so that testing at those interaction points is especially informative. Phase B5 completes the work by converting these perspectives into a unified test protocol: specific questions, organized for usability by commitment and by governance timing, that together confirm correct instantiation.

The governance implication is direct. At the close of Phase B5, governance of a Paper 2 deployment has four operationally distinct capabilities: it can specify what the deployment should look like (B2); it can recognize violation when the deployment drifts (B3); it can identify which dependencies are load-bearing when evaluating a proposed change (B4); and it can test whether the deployment is correctly instantiated at any point in its lifecycle (B5). These four capabilities together constitute a complete governance specification in the full sense: governance knows what, can recognize bad, understands dependencies, and can verify.

4. Phase B5's position in the Series B architecture

Series B is organized in six phases, each addressing a distinct governance dimension of Paper 2's architecture.

Phase B1 — Foundational commitments (20 notes). Phase B1 established the architectural vocabulary: the twenty commitments that Paper 2 introduces or extends. These commitments are the substrate of every subsequent phase. Phase B1 notes are referenced in every B2, B3, B4, B5, and B6 note as the source of the commitment being decomposed, violated, composed, or tested.

Phase B2 — Operational decompositions (110 notes). Phase B2 established the operational content of each commitment: what it means for a deployment to instantiate the commitment, expressed as behavioral requirements rather than design principles. Phase B2 is the largest phase in Series B because operational decomposition is the work most amenable to large-scale derivation: each commitment has multiple operational requirements, each requirement has multiple deployment variants, and each variant is independently publishable as a distinct prior-art contribution.

Phase B3 — Anti-pattern formalizations (30 notes). Phase B3 established the violation vocabulary: the recognizable failure signatures that identify when a commitment is not being honored. Phase B3 notes are organizationally complementary to Phase B2 notes — where B2 says what correct instantiation looks like, B3 says what incorrect instantiation looks like.

Phase B4 — Composition pairs (30 notes). Phase B4 established the dependency structure: the two-commitment interaction patterns whose correct handling requires simultaneous satisfaction of both commitments. Phase B4 notes address relationships between commitments rather than single commitments in isolation.

Phase B5 — Operational test notes (15 notes, this phase). Phase B5 established the test battery: the specific evidential questions that confirm correct instantiation across all commitments and all governance timings. Phase B5 is the phase that converts the specification from declarative to verifiable.

Phase B6 — Boundary cases (~15 notes, next phase). Phase B6 will establish the architecture's limits: the deployment configurations where Paper 2 commitments approach but do not exceed their governance boundaries. Phase B6 is described in §5 below.

5. Series B cumulative prior art at the close of Phase B5

At the close of Phase B5, Series B has produced 205 notes across its five completed phases:

Phase	Count	Governance dimension
B1 — Foundational commitments	20	What the commitments are
B2 — Operational decompositions	110	How they are operationally expressed
B3 — Anti-pattern formalizations	30	What violation looks like
B4 — Composition pairs	30	How commitments depend on each other
B5 — Operational test notes	15	How to verify correct instantiation
Series B total through B5	205	

The Series A prior-art record stands at 197 notes, derived from Paper 1 across its own six parallel phases (A1 through A6). Combined with the 205 Series B notes, the complete derivation-note series has produced **402 notes of public prior art** as of this note.

The 402-note total represents the public record of derivations from the two CKS theory papers as of Phase B5's close. Every note is independently deposited, independently dated, and independently citable as prior art for the specific derivation it formalizes. The cumulative effect is a prior-art record that covers the Paper 1 and Paper 2 architectures at fine granularity across foundational commitments, operational decompositions, anti-patterns, composition pairs, and

operational tests — leaving limited room for a subsequent party to claim novelty within the architecture's core territory.

6. Phase B6 preview: boundary cases as prior art

Phase B6 will formalize the edges of the Paper 2 architecture — the deployment configurations where the architecture's commitments approach their governance limits — as approximately fifteen boundary case notes. Boundary cases are distinct from anti-patterns: an anti-pattern (B3) is a configuration that violates a commitment; a boundary case (B6) is a configuration that is architecturally compliant but operates at the outer limit of what the architecture governs.

Formalizing boundary cases matters for the prior-art record for the same reason formalizing core commitments does: if boundary configurations are not published, a subsequent party can describe one and claim it as novel without bumping into the prior-art chain. The boundary territory is, in practice, where the most commercially relevant configurations often live — minimum viable deployments, transition states, high-frequency operational regimes.

Several Phase B6 boundary cases are already defined by the architecture and can be described in advance.

Minimum viable cell architecture. What is the smallest configuration that satisfies Paper 2's three-level requirement (cell, aspect, Self)? A deployment with a single cell, a single aspect, and a Self that is architecturally present but organizationally minimal is boundary-compliant rather than non-compliant: it satisfies the structural requirement at minimum cardinality. Formalizing this configuration as prior art prevents a subsequent claim that a single-cell Paper 2 deployment is a distinct invention.

Transition-period dual-level entity. What are the governance requirements when an entity performs functions at two structural levels simultaneously — for example, during a reorganization in which a cell is being elevated to aspect status before its prior cell role is retired? Paper 2 does not prohibit this configuration; it names death as a governed primitive (§6.4) and distinguishes functional obsolescence from capability supersession. A transition-period note formalizes the governance requirements that apply during the overlapping window, preventing the configuration from being claimed as novel.

High-frequency evolution events. What are the governance properties of a deployment in which mutation events in the DNA layer occur at frequencies that approach continuous rather than episodic change? Paper 2's instinct evolution mechanism (§7.2) does not specify a minimum interval between mutations; a high-frequency regime is architecturally compliant. Formalizing its governance requirements — specifically, what distinguishes high-frequency governed DNA evolution from ungoverned instinct drift — establishes this territory as prior art.

Aspect dissolution governance. What distinguishes aspect dissolution (a governed death at the aspect level) from aspect reorganization (a governed structural change that leaves the

aspect's cells intact under a new aspect definition)? Paper 2 names death as a governed primitive with two distinct types (§6.4, §8.4); the aspect-level analogue raises boundary questions about when dissolution governance applies. Formalizing this distinction prevents the claim that aspect-level lifecycle governance is distinct from cell-level lifecycle governance in a way that Paper 2's architecture does not already cover.

Self coherence under component replacement. The enterprise-brain Self (§9) is defined as architecturally coherent: its aspects and cells maintain relational integrity across the full Self. A boundary case arises when a significant fraction of the Self's cells are replaced within a short interval — either through mating, death, or rapid birth. Formalizing the governance requirements for Self coherence under high-turnover conditions establishes this configuration as prior art.

These examples are illustrative, not exhaustive. Phase B6 will formalize each boundary case with the same rigor as prior phases: traceable derivation from Paper 2 section numbers, no introduction of new axioms, operational test questions where applicable.

7. Conclusion: what Phase B5 accomplished and what comes next

Phase B5 began with a well-specified but unverifiable governance architecture — well-specified because Phases B1 through B4 had established what Paper 2's commitments are, how they decompose, what violations look like, and how they depend on each other; unverifiable because no phase had yet answered the question of how governance confirms that a live deployment actually instantiates those commitments. Phase B5 answered that question. Its fifteen notes produced a test library of approximately forty-one specific test questions, organized by commitment (B5.02–B5.09) and by governance timing (B5.10–B5.13), with a coverage map demonstrating completeness (B5.14).

The result is a governance specification in the fullest operational sense. Governance of a Paper 2 deployment now has a complete evidence-gathering protocol: it can identify what to build (B2), recognize divergence (B3), trace load-bearing dependencies (B4), and confirm correct instantiation at any lifecycle moment (B5). The architecture is not only defensible as a design — it is governable as an operation.

Phase B6 extends the prior-art record in the remaining direction: the boundary territory where Paper 2 commitments approach their limits. Its approximately fifteen notes will formalize minimum viable configurations, transition states, high-frequency regimes, and other architectural edge conditions, ensuring that the deployment configurations most likely to attract commercial interest are protected as prior art before they can be claimed as novel.

At the close of Phase B5, Series B stands at 205 notes; the complete series stands at 402. The boundary case territory is the final prior-art frontier for Series B.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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